The Relativistic Point Particle

Proseminar Spring 2023 - Introduction to String theory[†]

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Abstract. This report is primarily based on Barton Zwiebach's "A first course in String theory" from 2009 [Zwi09]. First, relevant aspects from special relativity and Lagrangian mechanics are summarized to find the action for a free relativistic point particle using an intuitive approach. Second, the action's reparameterization invariance is introduced, followed by the derivation of the Lorentz invariant equations of motion describing the dynamics of the free relativistic point particle using Hamilton's principle. Finally, the cases of a relativistic point particle with electric charge and a relativistic particle in curved space are briefly investigated.

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Contents

1 Introduction **2**

2 Special relativity **3**

3 Lagrangian mechanics **5**

4 Action for a relativistic point particle **7**

5 Reparameterization invariance **10**

6 Equations of motion **11**

7 Relativistic particle with electric charge **13**

8 Relativistic particle in curved space **14**

References **15**

1 Introduction

The relativistic point particle is a simple object that moves through spacetime under the influence of forces. It is described by classical mechanics, which was developed by Isaac Newton and later extended by Albert Einstein's theory of relativity.

String theory, on the other hand, proposes that the universe's fundamental building blocks are not point-like particles but rather one-dimensional objects called strings. These strings vibrate in various modes, which give rise to the different types of particles we observe.

Despite the differences between point particles and strings, there are many similarities in the mathematics describing their behavior. Later in the course, we will see that the mathematics of a free string is very similar to that of a free relativistic point particle. Studying the relativistic point particle, therefore, provides a valuable starting point for understanding some of the basic concepts of String theory.

The principles of special relativity that describe the behavior of relativistic point particles can be extended to describe the behavior of strings in a way consistent with the theory of general relativity.

2 Special relativity

This section is mainly based on chapters 2.1-2.2 (pp. 13-15) from Zwiebach's book [Zwi09].

Special relativity is a theory by Albert Einstein to describe the physical laws governing the behavior of objects moving at high speeds relative to one another.

The theory is based on two postulates: the physical laws have the same form in all inertial reference frames, and the speed of light in a vacuum is constant and independent of the source's and observer's motion.

We represent the motion of a relativistic particle with a path in spacetime called the particle's world-line

$$x^\mu = [x^0, x^1, x^2, x^3] \equiv [ct, x, y, z].$$

The two postulates imply that two observers in different Lorentz frames S and S' may disagree on the spacetime coordinates of events, but they will agree on the value of the invariant interval Δs^2 given by

$$-\Delta s^2 \equiv -(\Delta x^0)^2 + (\Delta x^1)^2 + (\Delta x^2)^2 + (\Delta x^3)^2 = -\Delta s'^2,$$

where $\Delta t = \Delta x^0/c$ and $\Delta x = [\Delta x^1, \Delta x^2, \Delta x^3]$ are the time- and position differences between the two separated events.

If $\Delta s^2 > 0$, we call the separation timelike. In this case, we define the proper time $\Delta t_p = \sqrt{\Delta s^2}/c$ as the time difference measured by an observer who observes the events happening at the same position. If $\Delta s^2 = 0$, we call the separation lightlike. If $\Delta s^2 < 0$, we call the separation spacelike.

In order to simplify, we introduce a new notation including infinitesimal coordinate differences dx^μ. This yields

$$-ds^2 = -(dx^0)^2 + (dx^1)^2 + (dx^2)^2 + (dx^3)^2.$$

Further simplification of the following form

$$dx_\mu = [dx_0, dx_1, dx_2, dx_3] \equiv [-dx^0, dx^1, dx^2, dx^3]$$

then yields

$$-ds^2 = \sum_{\mu=0}^{3} dx_\mu dx^\mu = dx_\mu dx^\mu,$$

where we summed over indices that appear twice in a single term (Einsteins summation convention). We can simplify this even further by introducing the Minkowski metric that describes the geometry of spacetime

$$-ds^2 = \eta_{\mu\nu} dx^\mu dx^\nu,$$

where

$$\eta_{\mu\nu} \equiv \begin{bmatrix} -1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \quad dx_\mu = \eta_{\mu\nu} dx^\nu.$$

The relations between coordinates of one event in two different inertial frames S and S' are called Lorentz transformations. Assuming that the two frames have parallel axes and that their origins coincide at the same time $t = t' = 0$, a boost of S' along the x^1 direction with constant velocity $\beta \equiv v/c$ is given by

$$\begin{aligned} x'^0 &= \gamma(x^0 - \beta x^1), \\ x'^1 &= \gamma(-\beta x^0 + x^1), \\ x'^2 &= x^2, \\ x'^3 &= x^3, \end{aligned}$$

where $\gamma \equiv 1/\sqrt{1-\beta^2} = 1/\sqrt{1 - v^2/c^2}$.

In general, we write a Lorentz transformation L as the linear relation

$$x'^\mu = L^\mu_\nu x^\nu.$$

In the case of the boost above, we can therefore write

$$[L] = L^\mu_\nu = \begin{bmatrix} \gamma & -\gamma\beta & 0 & 0 \\ -\gamma\beta & \gamma & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}.$$

3 Lagrangian mechanics

This section is mainly based on chapter 4.5 (pp. 78-81) from Zwiebach's book [Zwi09].

The Lagrangian of a system is defined by

$$L = T - V,$$

where T is the system's kinetic energy, and V is the system's potential energy. In the case of a point particle of mass m moving in the x direction under the influence of a time-independent potential $V(x)$, the non-relativistic Lagrangian is given by

$$L(t) = \frac{1}{2}m(\dot{x}(t))^2 - V(x(t)), \quad \dot{x}(t) \equiv \frac{dx(t)}{dt}.$$

The above Lagrangian only depends implicitly on time because it is a function of the position and its time derivative but not explicitly a function of time itself.

The action S is a functional taking a function of time and space as input and returning a scalar. It is defined as

$$S[x] = \int_{\mathcal{P}} L(t)dt = \int_{t_i}^{t_f} \left[\frac{m}{2}(\dot{x}(t))^2 - V(x(t))\right] dt,$$

where $\mathcal{P}$ is a path $x(t)$ between an initial position x_i at an initial time t_i and final position x_f at a final time $t_f > t_i$.

Hamilton's principle states that the true path of a system $\mathcal{P}$ between two specified endpoints in space and time is the one that minimizes the above action integral. In other words, Hamilton's principle states that the true path of a system is the one that makes the action stationary, meaning that a slight variation of the path should not change the action to the first order.

To obtain the equations of motion, we need to vary the action integral S with respect to position $x(t)$ while holding the endpoints fixed. This is done by considering a slight variation $\delta x(t)$ to the path followed by the system, such that

$$\delta x(t_i) = \delta x(t_f) = 0$$

at the endpoints. The variation $\delta x(t)$ leads to a corresponding variation in the

Lagrangian and, therefore, to a variation in the action integral

$$\begin{aligned} S[x+\delta x] &= \int_{t_i}^{t_f} \left[\frac{m}{2} \left(\frac{d}{dt}(x(t)+\delta x(t)) \right)^2 - V(x(t)+\delta x(t)) \right] dt \\ &= S[x] + \int_{t_i}^{t_f} \left[m\dot{x}(t)\frac{d}{dt}\delta x(t) - V'(x(t))\delta x(t) \right] dt + \mathcal{O}((\delta x)^2) \\ &\equiv S + \delta S, \end{aligned}$$

where we Taylor-expanded the potential $V(x(t))$, ignoring terms of order $(\delta x)^2$. One can then see that

$$\begin{aligned} \delta S &= \int_{t_i}^{t_f} \left[m\dot{x}(t)\frac{d}{dt}\delta x(t) - V'(x(t))\delta x(t) \right] dt \\ &= \int_{t_i}^{t_f} \delta x(t)(-m\ddot{x}(t) - V'(x(t)))dt, \end{aligned}$$

where we integrated by parts such that no derivatives act on $\delta x(t)$.

The action is stationary if δS vanishes for every $\delta x(t)$. This yields Newton's second law for a particle in a time-independent potential $V(x)$ and, therefore, the expected equation of motion given by

$$m\ddot{x}(t) = -V'(x(t)).$$

4 Action for a relativistic point particle

The following four sections are mainly based on the corresponding chapters 5.1-5.4 (pp. 89-98) from Zwiebach's book [Zwi09].

We ultimately describe the dynamics of the relativistic point particle with the equations of motion. Looking at the action first will provide a more complete, elegant, and convenient description of the system's dynamics than the equations of motion because they can become quite complex.

Let us first look at the free non-relativistic point particle. As explained in section (3), for a dynamical system, the action is obtained by integrating the Lagrangian over time. We find

$$S_{\text{nr}} = \int L_{\text{nr}} dt = \int \frac{m}{2} v^2(t) dt, \quad v^2 \equiv \vec{v} \cdot \vec{v}, \quad \vec{v} = \frac{d\vec{x}}{dt}, \quad v = |\vec{v}|.$$

Hamilton's principle yields the equation of motion

$$\frac{d\vec{v}}{dt} = 0,$$

i.e., the free non-relativistic point particle moves with constant velocity. This can also be derived directly from Newton's second law

$$\vec{F} = m\frac{d\vec{v}}{dt} \Rightarrow \frac{d\vec{v}}{dt} = 0.$$

We expect the free relativistic point particle to also move with constant velocity, therefore, one might assume that S_{nr} also holds in the relativistic case. This is wrong! The non-relativistic action allows for any constant velocity of the particle, even one faster than the speed of light. On top of that, the equations of motion obtained from S_{nr} are not Lorentz invariant which we require for relativistic particles. Due to this, we have to construct a new action S for the free relativistic point particle.

First, we expect S to yield the non-relativistic action S_{nr} in the non-relativistic limit. On top of that, as stated in section (2), we expect the equations of motion obtained from the action S to be Lorentz invariant. This means that they have the same form in all frames. How does this constrain the form of S? Suppose one observer states that for a specific world-line, the action is stationary. Then this world-line satisfies the equation of motion in that frame. If all observers now agree on the value of the action, they agree that the action is stationary for this world-line,

meaning that it satisfies the equations of motion in all frames. This means that all observers must agree on the value of S.

The non-relativistic action S_{nr} is not invariant under a Galilean boost which is a symmetry of corresponding equations of motion. Is Lorentz invariance then a too strong constraint on the relativistic action? Fortunately, this constraint is not a problem.

The proper time ds/c of the particle will be the same in all frames. As we have seen above, we want all observers to compute the same S. We, therefore, expect the action S to be proportional to the proper time, i.e.,

$$S \propto \int_{\mathcal{P}} \frac{ds}{c}.$$

Since the Lagrangian has units of energy, the action must have units of energy times time

$$[S] = M\frac{L^2}{T^2}T = \frac{ML^2}{T}.$$

Proper time has units of time, meaning that we need an additional factor with units of energy or mass times velocity squared. To preserve the Lorentz invariance of ds/c, the components of the additional factor must also be Lorentz invariant. We know that the rest mass m of the particle and the speed of light are Lorentz invariant. Therefore, we write

$$S \propto mc^2 \int_{\mathcal{P}} \frac{ds}{c} = mc \int_{\mathcal{P}} ds.$$

As we will see below, showing that this action matches S_{nr} in the non-relativistic limit yields a missing minus sign, but the unit coefficient is correct

$$S = -mc \int_{\mathcal{P}} ds.$$

From section (2) we know that we can relate the infinitesimal time dt to the infinitesimal invariant interval ds by

$$ds = cdt\sqrt{1 - \frac{v^2}{c^2}}.$$

This only holds for $v \leq c$. Therefore the maximum speed constraint is already built

into our action S as required. We find that

$$S = -mc^2 \int_{t_i}^{t_f} dt \sqrt{1 - \frac{v^2}{c^2}}$$

and the corresponding Lagrangian is then given by

$$L = -mc^2 \sqrt{1 - \frac{v^2}{c^2}}.$$

Taylor expanding the Lagrangian under the assumption that $v \ll c$ (i.e. expansion around $v/c = 0$) yields

$$L \approx -mc^2 \left(1 - \frac{v^2}{2c^2}\right) = -mc^2 + \frac{mv^2}{2}.$$

The first term is constant and irrelevant to the equations of motion. It can therefore be ignored.

Indeed, our relativistic Lagrangian yields the physics we know in the form of the canonical momentum

$$\vec{p} = \frac{\partial L}{\partial \vec{v}} = -mc^2 \left(-\frac{\vec{v}}{c^2}\right) \frac{1}{\sqrt{1 - v^2/c^2}} = \frac{m\vec{v}}{\sqrt{1 - v^2/c^2}} = \gamma m \vec{v}$$

and the Hamiltonian

$$\mathcal{H} = \vec{p} \cdot \vec{v} - L = \frac{m\vec{v}}{\sqrt{1 - v^2/c^2}} + mc^2 \sqrt{1 - v^2/c^2} = \frac{mc^2}{\sqrt{1 - v^2/c^2}} = \gamma mc^2.$$

5 Reparameterization invariance

To explicitly calculate S, one would have to parameterize the particle's world-line $\mathcal{P}$. An important property of the relativistic point particle action S is that it is invariant under the reparameterization of the particle's world-line, i.e., S does not change with the parameterization chosen to calculate it.

Usually, we describe the position as a function of time. However, in the relativistic case, we want to treat space and time equally and therefore have to parameterize space and time in terms of a new parameter $\tau \in [\tau_i, \tau_f]$ that is strictly increasing as the world-line goes from the initial point $x_i^\mu \equiv x^\mu(\tau_i)$ to the final point $x_f^\mu \equiv x^\mu(\tau_f)$.

We can rewrite the invariant interval ds^2 as

$$ds^2 = -\eta_{\mu\nu} dx^\mu dx^\nu = -\eta_{\mu\nu} \frac{dx^\mu}{d\tau} \frac{dx^\nu}{d\tau} (d\tau)^2.$$

We are only interested in world-lines for which $v \leq c$, i.e., $ds^2 > 0$ for every τ. Therefore we can write

$$S = -mc \int_{\tau_i}^{\tau_f} \sqrt{-\eta_{\mu\nu} \frac{dx^\mu}{d\tau} \frac{dx^\nu}{d\tau}} d\tau.$$

In order to see the reparameterization invariance, we change from τ to τ' with $d\tau' = \frac{\partial \tau'}{\partial \tau} d\tau$. By the chain rule, we have that

$$\frac{dx^\mu}{d\tau} = \frac{dx^\mu}{d\tau'} \frac{d\tau'}{d\tau}$$

from which we find that

$$\begin{aligned} S &= -mc \int_{\tau_i}^{\tau_f} \sqrt{-\eta_{\mu\nu} \frac{dx^\mu}{d\tau'} \frac{dx^\nu}{d\tau'} \left(\frac{d\tau'}{d\tau}\right)^2} d\tau \\ &= -mc \int_{\tau_i}^{\tau_f} \sqrt{-\eta_{\mu\nu} \frac{dx^\mu}{d\tau'} \frac{dx^\nu}{d\tau'}} \left|\frac{d\tau'}{d\tau}\right| d\tau \\ &= -mc \int_{\tau_i'}^{\tau_f'} \sqrt{-\eta_{\mu\nu} \frac{dx^\mu}{d\tau'} \frac{dx^\nu}{d\tau'}} d\tau' \\ &= S', \end{aligned}$$

reparameterization invariance of the relativistic action S.

6 Equations of motion

As we have seen in section (3), in order to find the equations of motion from the action, we must calculate the variation δS for a small variation of the world-line by $\delta x^\mu(\tau)$.

For the relativistic point particle, we have

$$\delta S = -mc \int \delta(ds).$$

We can find the variation $\delta(ds)$ from the simpler variation of $(ds)^2$

$$2ds\delta(ds) = -2\eta_{\mu\nu}\delta\left(\frac{dx^\mu}{d\tau}\right)\frac{dx^\nu}{d\tau}(d\tau)^2,$$

where we must vary with $\frac{dx^\mu}{d\tau}$ and $\frac{dx^\nu}{d\tau}$, but since $\eta_{\mu\nu}$ is symmetric, we can only vary $\frac{dx^\mu}{d\tau}$ and add a factor of 2. Using that the variation of velocity is the same as the time derivative of the variation of position, we find

$$\delta(ds) = -\eta_{\mu\nu}\frac{d(\delta x^\mu)}{d\tau}\frac{dx^\nu}{ds}d\tau = -\frac{d(\delta x^\mu)}{d\tau}\frac{dx_\mu}{ds}d\tau,$$

where $\eta_{\mu\nu}dx^\nu = dx_\mu$.

From this, we find the variation of the action

$$\delta S = mc\int_{\tau_i}^{\tau_f}\frac{d(\delta x^\mu)}{d\tau}\frac{dx_\mu}{ds}d\tau.$$

Using that $p_\mu = mc\frac{dx_\mu}{ds}$, we can simplify

$$\delta S = mc\int_{\tau_i}^{\tau_f}\frac{d(\delta x^\mu)}{d\tau}p_\mu d\tau.$$

Again, as we have already seen in section (3), we need to write the variation δS such that we have δx^μ multiplying an object under the integral to retrieve the equations of motion. We know that

$$\frac{d(\delta x^\mu)}{d\tau}p_\mu = \frac{d}{d\tau}\left(\delta x^\mu p_\mu\right) - \delta x^\mu(\tau)\frac{dp_\mu}{d\tau},$$

therefore

$$\delta S = \int_{\tau_i}^{\tau_f} d\tau \frac{d}{d\tau} \left(\delta x^\mu p_\mu\right) - \int_{\tau_i}^{\tau_f} d\tau \delta x^\mu(\tau) \frac{dp_\mu}{d\tau},$$

where the first integral yields $\delta x^\mu p_\mu$ evaluated at the endpoints of the world-line. This integral vanishes because $\delta x^\mu(\tau_i) = \delta x^\mu(\tau_f) = 0$.

We obtain the equation of motion by demanding that the second term must vanish for arbitrary $\delta x^\mu(\tau)$

$$\frac{dp_\mu}{d\tau} = 0,$$

or equivalently $\frac{dp^\mu}{d\tau} = 0$. This implies that the momentum of the point particle is constant along the world-line for any arbitrary parameter τ.

If we choose to parameterize the world-line with the proper time s, we get

$$\frac{d}{d\tau} p_\mu = \frac{d}{d\tau}\left(mc\frac{dx_\mu}{ds}\right) = \frac{d}{ds}\left(mc\frac{dx^\mu}{ds}\right) = mc\frac{d^2x^\mu}{ds^2} = 0.$$

This shows that the non-relativistic equation of motion only holds for the proper time and not an arbitrary parameter.

The above equation of motion is indeed Lorentz invariant, as substituting $x'^\mu = L^\mu_\nu x^\nu$ yields

$$0 = \frac{d^2x'^\mu}{ds^2} = \frac{d^2(L^\mu_\nu x^\nu)}{ds^2} = L^\mu_\nu \frac{d^2x^\nu}{ds^2}.$$

Since L is invertible, this implies that

$$\frac{d^2x^\nu}{ds^2} = 0,$$

i.e., Lorentz invariance.

7 Relativistic particle with electric charge

Suppose a point particle has an electrical charge q and electromagnetic fields are present. In that case, the particle will not be free anymore, i.e., it will be subject to external forces, and its four-momentum will not be constant anymore.

Similar to the derivation of the action in section (4), we try to find an action that yields the following known equation of motion upon variation

$$\frac{dp_\mu}{ds} = \frac{q}{c} F_{\mu\nu} \frac{dx^\nu}{ds},$$

where $F_{\mu\nu}$ is the field strength. We can see that ds appears on both sides. Therefore we can rewrite the equation with the general parameter τ

$$\frac{dp_\mu}{d\tau} = \frac{q}{c} F_{\mu\nu} \frac{dx^\nu}{d\tau}.$$

The interaction with the fields suggests adding another Lorentz invariant term to the action involving the particle's four-velocity. We find

$$\begin{aligned} S &= -mc \int_{\mathcal{P}} ds + \frac{q}{c} \int_{\mathcal{P}} d\tau A_\mu(x(\tau)) \frac{dx^\mu}{d\tau} \\ &= -mc \int_{\mathcal{P}} ds + \frac{q}{c} \int_{\mathcal{P}} A_\mu(x) dx^\mu, \end{aligned}$$

where A_μ is the gauge potential. This action is Lorentz invariant because the product $A_\mu(x)dx^\mu$ is a Lorentz scalar. We then obtain the equation of motion by setting to zero the variation of S under a change δx^μ of the particle world-line

$$\frac{dp_\mu}{d\tau} = \frac{q}{c} F_{\mu\nu} \frac{dx^\nu}{d\tau}.$$

8 Relativistic particle in curved space

In a curved space, we write the invariant interval as

$$-ds^2 = g_{\mu\nu}(x)dx^\mu dx^\nu,$$

where $g_{\mu\nu}(x)$ is the position-dependent curved space metric.

We find that the action has the same form as the action in the non-curved case

$$S = -mc\int_{\mathcal{P}} ds = -mc\int_{\tau_i}^{\tau_f}\sqrt{-g_{\mu\nu}(x)\frac{dx^\mu}{d\tau}\frac{dx^\nu}{d\tau}}d\tau.$$

By setting to zero the variation of S under a variation δx^μ of the particle world-line parameterized with the proper time s, we obtain the equation

$$\frac{d}{ds}\left[g_{\mu\rho}\frac{dx^\mu}{ds}\right] = \frac{1}{2}\frac{\partial g_{\mu\nu}}{\partial x^\rho}\frac{dx^\mu}{ds}\frac{dx^\nu}{ds}$$

from which we can derive the geodesic equation

$$\frac{d^2x^\mu}{ds^2} = -\Gamma^\mu_{\nu\rho}\frac{dx^\nu}{ds}\frac{dx^\rho}{ds},$$

where

$$\Gamma^\mu_{\nu\rho} = \frac{1}{2}g^{\mu\sigma}\left(\frac{\partial g_{\nu\sigma}}{\partial x^\rho} + \frac{\partial g_{\rho\sigma}}{\partial x^\nu} - \frac{\partial g_{\nu\rho}}{\partial x^\sigma}\right)$$

are the Christoffel symbols. The geodesic equation is the general relativity analog to the equations of motion in classical mechanics.

The lecture notes on general relativity by Matthias Blau [Bla22] include a derivation of the geodesic equation from the extremisation of proper time (Chapter 2.3, pp. 56-58) and a calculation of the Christoffel symbols (Chapter 3.1, p.79).

References

[Bla22] Matthias Blau. *Lecture Notes on General Relativity*. Nov. 2022. URL: `http://www.blau.itp.unibe.ch/newlecturesGR.pdf` (visited on 05/03/2023).

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